

Xuran Cai

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Research Interests

I am generally interested in Theoretical Computer Science (TCS), with a focus on graph theory and its algorithmic and structural aspects, which includes

- Program verification and formal methods
- SAT/SMT solving and constraint solving techniques
- Compiler optimization and program analysis
- Applications of theoretical methods to blockchain systems and distributed computation

Education

Oxford-Max-Plank joint Program, Doctor of Philosophy Sep 2025 - Present
German & UK

- Joint Program by University of Oxford and Max Plank Institute
- Research Interest: Graph Theory, Verification, general TCS
- Supervisor: Prof. Amir Goharshady(Oxford) and Prof. Rupak Majumdar(MPI)

Hong Kong University of Science and Technology, Master of Philosophy Sept 2023 – June 2025
Hong Kong SAR

- Thesis: Enhancing Compiler Optimization Efficiency through Grammatical Decompositions of Control-Flow Graphs
- Supervisor: Prof. Amir Goharshady and Prof. Jiasi Shen

University of Wisconsin-Madison, Bachelor of Science Sept 2019 – May 2023
USA

- GPA: 3.86/4.0
- Major: Double Major in Computer Science and Mathematics
- Graduate with Distinction and Major Distinction in CS

Publications

Extending Programmable Blockchain Protocols to Enable and Incentivize Parallel Execution of Smart Contracts Sep 2026

T. Barakbayeva, X. Cai, Z. Cai, A.K. Goharshady

International Conference on Mathematical Research for Blockchain Economy, MARBLE'26

Polynomial Invariant Generation for Floating Point Programs July 2026

X. Cai, L. Chen, H. Fu

International Conference on Computer Aided Verification, CAV'26

Series-Parallel-Loop Decompositions of Control-flow Graphs February 2026

X. Cai, A.K. Goharshady, S. Hitarth, C.K. Lam

Journal of Systems Architecture, JSA'26

Efficient Algorithms for Partial Constraint Satisfaction Problems over Control-flow Graphs December 2025

X. Cai, A.K. Goharshady

International Symposium on Software Engineering: Theories, Tools, and Applications, SETTA'25

Enhancing Compiler Optimization Efficiency through Grammatical Decompositions of Control-Flow Graphs July 2025

X. Cai

Mphil Final Thesis

Faster Chaitin-like Register Allocation via Grammatical Decompositions of Control-Flow Graphs

March 2025

X. *Cai*, A.K. Goharshady, S. Hitarth, C.K. Lam

International Conference on Architectural Support for Programming Languages and Operating Systems, ASPLOS'25

Faster Lifetime-optimal Speculative Partial Redundancy Elimination for Goto-free Programs

November 2024

X. *Cai*, A.K. Goharshady

International Symposium on Software Engineering: Theories, Tools, and Applications, SETTA'24

Teaching Assistant Experiences

Concurrent Programming

MT Semester 2026-27

- Institute: Oxford
- Instructor: prof. Vincent Cheval

Design and Analysis of Algorithms

HT Semester 2025-26

- Institute: Oxford
- Instructor: prof. Sandra Kiefer

Functional Programming

MT Semester 2025-26

- Institute: Oxford
- Instructor: prof. Andrzej Murawski

Honors Discrete Mathematical Tools for Computer Science

Fall Semester 2024-25

- Institute: HKUST
- Instructor: prof. Amir Goharshady

Discrete Mathematical Tools for Computer Science

Spring Semester 2023-24

- Institute: HKUST
- Instructor: prof. Sunil Arya and prof. Jiasi Shen

Introduction to Computer Networks

Spring Semester 2021-22

- Institute: UW-Madison
- Instructor: prof. Suman Banerjee

Funding

Researcher Salary – MPI-SWS, Kaiserslautern, German

Jan 2026 - present

Department Scholarship – University of Oxford, Oxford, UK

Oct 2025 - present

Postgraduate Scholarship – HKUST, Hong Kong SAR

Sep 2023 - July 2025

Internship

Summer Intern Software Developer, Intel – Shanghai, China

June 2021 - Aug 2021

- Worked on designing a service mesh structure with SGX hardware
- Designed and implemented the stereotype of the system

Conference/Summer Schools

HUAWEI 2012 Lab Summer School , Guangdong, China

Aug 17 - Aug 21, 2026

CAV'26 , Lisbon, Portugal

July 26 - July 29, 2026

Setta'25 , Oxford, UK

Dec 1 - Dec 3, 2025

Setta'24 , Hong Kong SAR

Nov 26 - Nov 28, 2024

FM'24 , Milan, Italy

Sep 9 - Sep 13, 2024

Summer School on Discrete Mathematics, Prague, Czech Republic

July 1- July 5, 2024